Dr. Silvia-Laura Pintea



Computer Vision Researcher
Nationality: Dutch | Pronouns: she/her/hers.
Roles: Researcher, lecturer, mentor, invited speaker, computer scientist.
Email: Silvia[dot]Laura[dot]Pintea[at]gmail[dot]com
Webpage: http://silvialaurapintea.github.io

My work is in the field of AI, specifically Computer Vision and Deep Learning, where I have extensive experience with video and image analysis, specifically scale-invariance, video motion representations and regression problems.
My current focus is on effective video analysis and robust motion representations.

Work Experience

Assistant prof.	**Department of Intelligent Systems (DIS), TiU, Tilburg, NL**
Job Description	Physics priors for accurate and computationally efficient video and motion analysis.
Period	June 2025 — Present
Postdoc (Guest researcher)	**Division of Image Processing (LKEB), LUMC, Leiden, NL** **Vision Lab, Delft University of Technology (TUDelft), Delft, NL**
Job Description	Efficient video and motion representations towards surgery video analysis.
Period	May 2022 — April 2025
Researcher	**Vision Lab, Delft University of Technology (TUDelft), Delft, NL**
Job Description	Focus on Deep Learning scale-invariance and efficiency.
Period	July 2020 — February 2022
Postdoc	**Vision Lab, Delft University of Technology (TUDelft), Delft, NL**
Job Description	Collaboration with LUMC and the Geophysics department at TUDelft. Working on image and video analysis: geometric priors, scale-invariance, video representations, etc.
Period	July 2016 — July 2020
R&D Engineer	**Layar/Blippar, Amsterdam, NL** (augmented reality company)
Job Description	Focus on efficient large-scale image matching and retrieval, using C++.
Period	January — June 2016

Education

PhD	**Computer Vision**, University of Amsterdam (UvA), Amsterdam, NL
Focus	Motion prediction, object localization, video representation learning
Thesis Title	Continuous Learning in Computer Vision
Period	2011 – 2015
MSc	**Artificial Intelligence**, University of Amsterdam, (UvA) Amsterdam, NL
Focus	Machine Learning, Computer Vision, Neural Networks, Game Theory.
Period | Avg. Grade	2009 – 2011 | 8.31 (out of 10)
BSc	**Computer Science**, University of Bucharest, Faculty of Mathematics and Computer Science, Bucharest, RO
Focus	Algorithms and data structures, Object oriented programming, Formal methods.
Period | Avg. grade	2005 – 2008 | 9.40 (out of 10)

Teaching and guiding

Teaching	BSc Introduction to deep learning (TiU, 2025-2026), BSc Computer vision (TiU, 2025-2026), MSc Image analysis (TiU, 2025-2026), Mathematics for PM (TiU, 2025-2026) BSc Image processing (TUDelft, 2018-2022), MSc Computer vision by deep learning (TUDelft, 2018-2022)
PhD copromotor	Dr. Yancong Lin, Dr. Xin Liu
PhD adviser	Xiaotong Zhang, Faeze Gholamiankhah, Xin Liu, Yancong Lin, Vedran Vukotic (visting student), Abolfazi Nadi (visiting student)
MSc adviser	Tobias Stahl, Yichao Zhang, Chengqiu Zhang, Xilin Li, Yue Liu, Omar Hommos, Xiaoming Wen, Jian Zheng, Ziyu Bao, Yordan Dimitrov, Nikhil Saldanha, Ziyu Bao, Jimmy Vlekke, Frans de Boer, Douwe Hoonhout, Marijn Sluijs

Qualifications, training and skills

English proficiency	Level C2, "English Language Assessment for Academic Staff" (TiU), in 2025.
BKO	"University teaching qualifications" (TUDelft), completed in 2022.
PDP	"Tenure Track Personal Development Program" (TUDelft), completed in 2021.
Programming frameworks	Jax, PyTorch, TensorFlow, Python, Caffe, OpenCV
Programming languages	Python, C/C++, Torch, PHP, Java, Matlab; (No vibe coding)

Community service, awards & other

Member	IPN Equality Diversity and Inclusion Working Group
Invited speaker	Data Science Seminars (Radboud, 2026), Digital Society Institute Meet Up (Twente, 2026), Startup Village (Amsterdam, 2022), AIRLab (Delft, 2019), NucTech meeting (Delft, 2018), Delft-Leiden Deep Learning Seminars (Leiden, 2019)
Invited lecturer	Deep Learning MSc course TUDelft (2019)
Award	Outstanding Reviewer: CVPR (2025, 2022, 2021), ICCV (2021), ECCV (2022), BMVC(2022, 2021) Best paper award for the Workshop on Hands, ECCVw (2018)
International conferences	ICPR (2026), ICCV (2023,2021), CVPR (2021), ECCV (2020, 2016, 2014), ICIP (2018, 2016)
Local conferences/events	NCCV (2018 - 2023, 2026), Video Understanding Symposium (2023), NVPHBV (2017), ICT.Open (2016, 2012)
Reviewer (since 2012)	CVPR, ECCV/ICCV, BMVC, WACV, ICPR, TIP, NeurIPs, IJCV

Key publications

- **"Deep kernel video approximation for unsupervised action segmentation"** *SL Pintea*, J. Dijkstra, International Conference on Pattern Recognition (ICPR) 2026.
- **"A step towards understanding why classification helps regression"** *SL Pintea*, Y. Lin, J. Dijkstra, JC van Gemert, International Conference on Computer Vision (ICCV) 2023.
- **"Resolution learning in deep convolutional networks using scale-space theory"**, *SL Pintea*, N Tomen, SF Goes, M Loog, JC van Gemert, Transactions on Image Processing (TIP), 2021.
- **"No frame left behind: Full Video Action Recognition"**, X Liu, *SL Pintea*, FK Nejadasl, O Booij, JC van Gemert. Computer Vision and Pattern Recognition (CVPR), 2021.
- **"Video Acceleration Magnification"**, Y Zhang, *SL Pintea*, Jan C. van Gemert, Computer Vision and Pattern Recognition (CVPR), 2017.
- **"Déjà Vu: Motion Prediction in Static Images."**, *SL Pintea*, JC van Gemert, AWM Smeulders, European Conference on Computer Vision (ECCV), 2014.

Complete list of publications: https://scholar.google.nl/citations?user=shTkx9EAAAAJ&hl=en